

ARVIND RAMASWAMI (US CITIZEN)

Machine Learning, Optimization, and Algorithms

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 arvind-ramaswami  arvindr9

WORK EXPERIENCE

ML Intelligence Engineer – Contract

Mercor

 Oct 2024 - Present  San Francisco, CA

ML Research Intern

Genesis Therapeutics

 Feb 2023 - Aug 2023  Burlingame, CA

- Startup for ML + molecular simulation technologies for drug discovery
- Built pose-ranking benchmarks to evaluate ligand pose scoring methods
- Developed ML-based scoring methods for ligand poses

ML Engineer Intern

PerfectRec

 Jun 2022 - Aug 2022  Seattle, Washington

- Developed the startup's first ML-powered recommendation system, using AWS Sagemaker.
- Built a framework that enabled fitting a custom scikit-learn model (including polynomial regression and gradient boosted machines) for the recommendation system.
- Conducted a literature review of learn-to-rank and proposed ways to apply learn-to-rank to product recommendations

Software Development Engineer Intern

Amazon

 May 2020 - Aug 2020  Seattle, Washington

- Designed a recommender system for the Amazon Advertiser Partner Network (AAPN), and outlined the AWS tools to be used for the recommender system.
- Performed exploratory data analysis on the advertiser dataset.

PUBLICATIONS

Publications

- J. Chen, L. Chen, Y. P. Liu, R. Peng, and A. Ramaswami, "Exponential convergence of sinkhorn under regularization scheduling," in *SIAM Conference on Applied and Computational Discrete Algorithms (ACDA23)*, SIAM, 2023, pp. 180–188.

Preprints / Submissions

- I. Attias, S. Hanneke, and A. Ramaswami, "Sample compression scheme reductions," *arXiv preprint arXiv:2410.13012*, 2024.

MOST PROUD OF



ICPC World Finalist

Placed Rank 6 in the 2022 ICPC North American Championship and went to the World Finals at Dhaka, Bangladesh



Meta Hacker Cup Top 200

Placed top 200 in Round 3 of the 2022 Meta Hacker Cup, out of over 27000 contestants worldwide

SKILLS

Python

C++

Algorithm Design

Machine Learning

Optimization

Pytorch

Cloud computing (AWS, Azure)

EDUCATION

PhD in Computer Science

Purdue University

 2023-Present

Specialization: Machine Learning / Artificial Intelligence

BS+MS in Computer Science

Georgia Institute of Technology

 2017 - 2022

Advisors: Richard Peng, Sebastian Pokutta, Jacob Abernethy

BS Threads: Theory and Intelligence

MS Specialization: Machine Learning

COURSEWORK

Math classes: Advanced Graph Theory, Graduate Probability I, Analysis I, Abstract Algebra I, Algebraic Topology I, High Dimensional Statistics

CS classes: Machine Learning Theory, Dynamic Algebraic Algorithms, High-Performance Computing, Deep Learning, Robot Intelligence: Planning, Applied Cryptography, Intro to Database Systems, NLP

Online certifications: Coursera Deep Learning Specialization

LEADERSHIP / SERVICE

Georgia Tech Programming Team: President

Georgia Institute of Technology

📅 Jan 2022 - Dec 2022

- Prepared content and lectured for weekly meetings in topics such as dynamic programming, data structures, and combinatorial optimization.
 - In charge of arranging team practices for ICPC.
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Big O Theory Club: President

Georgia Institute of Technology

📅 Aug 2020 - May 2021

- Gave talks about theoretical computer science
 - Examples of topics discussed: randomized algorithms, flows. Invited other students to give talks about their research.
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Other service

- Coach for Georgia ARML (American Regions Mathematics League) team: May - June 2022.
- Reviewer for Georgia Tech President's Undergraduate Research Awards (Fall 2022).

TEACHING

USA Computing Olympiad Instructor

X-Camp Academy

📅 January 2023 - Present

- Coach for the USACO Silver Prep Program
- Teaching 400-500 level classes – for students preparing for the USACO Silver / Gold Level -4 hours of teaching per week

Computer Science Instructor

Purdue University

📅 January 2024 - present

- Teaching an intermediate level class for competitive programming (CS 311, Competitive Programming)

Automata Teaching Assistant

Georgia Institute of Technology

📅 January 2021 - December 2022

- Teaching assistant under Dr. Zvi Galil's Automata and Complexity class (CS 4510) for four semesters.

Algorithms Teaching Assistant

Georgia Institute of Technology

📅 August 2019 - December 2019

- Teaching assistant for Professor Eric Vigoda's algorithms class (CS 3510).